



QUANTIE INSTITUTE OF ADVANCED STUDIES

(A Private Limited Company)

Certificate Program in Data Science

"Learn like a quant, think like a trader, build like an engineer"

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A rigorous, mathematics-driven programme that builds strong theoretical foundations and turns them into real-world analytical and AI solutions in Python, for learners aiming at serious careers in analytics, machine learning, quantitative research and AI-driven decision systems.

Curriculum

Module 0 | Foundations

Mathematical foundations, SQL and Python for data science.

Module 1 | Time Series Analysis

OLS & its mathematics, statistical & normality tests, WLS, AR/MA/ARIMA, cointegration, VAR, ARCH/GARCH, EWMA, Parkinson, Garman–Klass, Rogers–Satchell, Yang–Zhang.

Module 2 | Machine Learning

Regularised regression (Lasso, Ridge, Elastic Net), trees, SVM, ensembles, imbalanced classes, PCA/ICA/LDA/SOM, clustering, feature engineering, CV, pipelines, tuning.

Module 3 | Deep Learning

MLP, CNN, RNN, LSTM, GRU, Transformers; activation & loss functions, optimisers and hyper-parameter tuning.

Module 4 | Reinforcement Learning

MDPs & Bellman equations, value- and policy-based methods, RL for trading, rebalancing & execution, risk-aware RL, robustness under non-stationarity.

Module 5 | Advanced & Agentic AI

LLM & GenAI foundations, agent architectures for research workflows, AI with quant market signals, operational risk, governance & validation.

Module 6 | Capstone Project — an end-to-end applied project, mentored and reviewed

Career Outcomes

- ▶ Data Analyst
- ▶ Data Scientist
- ▶ Machine Learning Engineer
- ▶ Quantitative Analyst
- ▶ AI Research Scientist

Programme Fees

INR 1,00,000 Domestic
USD 2,500 International

20% DISCOUNT AVAILABLE

More Savings

- ▶ **Group:** extra 5% for 5+ enrolments, 10% for 10+.
- ▶ **Regional:** 50% off the international fee for African nationals and candidates from Nepal, Bhutan, Bangladesh, Sri Lanka, Maldives and Pakistan.

Apply Now

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